Matrix Theory I: Matrices, Trace and Determinants: Answer Key

Problem sheet for the Day 1 class · CSIR NET previous year questions

Answers

Q	1	2	3	4	5	6	7	8	9	10
Answer	(D)	(C)	(B)	(D)	(C)	(B)	(B)	(B)	(A),(C)	(A),(C)

Hints and short solutions

1. **(D)** CSIR NET June 2020 $\operatorname{tr} A^2 = \lambda_1^2 + \lambda_2^2 = (\lambda_1 + \lambda_2)^2 - 2\lambda_1\lambda_2 = 9 - 2 = 7$.
2. **(C)** CSIR NET Dec 2016 Transpose D_2, take -1 out of the first column and the second row, then swap two rows: one sign change survives. (Checked by direct expansion: $D_1 + D_2 = 0$.)
3. **(B)** CSIR NET Dec 2016 A^k is rotation by $k\theta$; $2015 = 31 \times 65$, so $2015\theta = 130\pi$ and $A^{2015} = I$.
4. **(D)** CSIR NET Dec 2015 An idempotent has eigenvalues 0 and 1; since $A \neq I$, 0 occurs. (A), (B), (C) are standard facts for idempotents.
5. **(C)** CSIR NET June 2018 For 2×2 matrices $\det(A + I) = \det A + \operatorname{tr} A + 1$.
6. **(B)** CSIR NET Dec 2014 $B(A + B + A^2) = I$, so B has a right inverse and is invertible. $A = 0, B = I$ kills (A) and (D).
7. **(B)** CSIR NET Dec 2014 Reversing $1, \ldots, n$ is a product of $\lfloor n/2 \rfloor$ transpositions.
8. **(B)** CSIR NET June 2016 $t \mapsto \det C(t)$ is a polynomial, negative at 0 and positive at 1: it vanishes somewhere in between (and only finitely often, so (C) fails; (D) fails too since it is non-zero at all but finitely many t).
9. **(A),(C)** CSIR NET Dec 2016 $P = -QPQ^{-1}$ gives $\operatorname{tr} P = -\operatorname{tr} P$, so $\operatorname{tr} P = 0$; likewise $\operatorname{tr} Q = 0$.
10. **(A),(C)** CSIR NET June 2019 $(c^{i+j}a_{ij}) = DAD$ with $D = \operatorname{diag}(c, c^2, c^3)$, so the determinant is $(\det D)^2 \det A = c^{12} \det A$; $c^{12} = 1$ for $c = -1$ and for $c = i$.